

MOHAMED HELMY

Software Engineer | Flutter & .NET Developer

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LinkedIn: linkedin.com/in/mohamedHelmy-dev | GitHub: github.com/MooHelmy | Portfolio: MooHelmy_Portfolio

PROFILE

Software Engineer with 3+ years of experience building scalable, cross-platform apps with Flutter and Dart, using Clean Architecture, MVVM, and modern state management. Recently expanded into backend development, shipping three ASP.NET Core Web API projects — e-commerce, flight booking, and hotel management — built with Entity Framework Core, SQL Server, ASP.NET Identity, and JWT authentication. Shipped an independent portfolio of AI-powered, offline-first apps — including on-device speech recognition — cutting manual workflows by up to 80% through automation.

CORE SKILLS

Languages: Dart, C#, Java, Kotlin, Python, C++, C, Assembly, SQL

Mobile Development: Material UI, Responsive UI, Flutter Widgets, Navigation, Async Programming

Frameworks & Tools: Flutter, Android SDK, Google Maps API, Stripe SDK, Dio, HTTP, Git, GitHub, GitHub Actions, CI/CD, Postman, Android Studio, VS Code

State Management: Bloc, Cubit, Provider, Multithreading

Architecture: Clean Architecture, MVVM, Repository Pattern, SOLID, OOP, Design Patterns

Backend & APIs: ASP.NET Core Web API, ASP.NET Identity, Entity Framework Core, JWT Authentication, REST APIs, JSON Parsing, Dependency Injection, Firebase (Auth, Firestore, Storage, App Distribution)

Database & Storage: SQL Server, Hive, MySQL, Oracle, Secure Storage

AI / ML & Speech: Machine Learning, CNN, Computer Vision, On-Device AI Model Integration, NLP, Speech Recognition

Testing & Security: Unit Testing, Widget Testing, Performance Profiling, Background Services, Role-Based Access Control

Additional Skills: IoT, Embedded Systems, Data Structures, Algorithms, Computer Architecture, Computer Networks, DBMS, Computer Graphics, Web Technologies, Data Science

PROFESSIONAL EXPERIENCE

Software Engineer – Military Service | Egyptian Armed Forces (MTC)

Dec 2024 – Dec 2025 | Cairo, Egypt

Engineered a secure Flutter system to digitize military educational logistics, lecture tracking, and attendance, fully replacing a paper-based workflow.

- Developed a Flutter application that automated attendance and reporting workflows, reducing manual data-entry time by approximately 80% and eliminating transcription errors.
- Architected an offline-first sync engine using Hive local storage, ensuring zero data loss in restricted or low-connectivity network environments.
- Automated multi-step reporting pipelines, cutting report generation from several hours to under 5 minutes while improving data accuracy for senior staff.
- Implemented AES-based secure storage and role-based access control for handling sensitive military data, meeting internal security compliance standards.

Flutter Developer | Freelancer (Self-Employed)

Dec 2024 (Part-Time) – Present | Remote

- Designed and delivered 3+ cross-platform Flutter applications using Clean Architecture, MVVM, and Bloc/Cubit state management for diverse client requirements.
- Integrated REST APIs with JSON parsing, Firebase Auth, Firestore, and third-party SDKs including Stripe payment processing.
- Applied performance profiling and optimization techniques to maintain smooth 60fps UI rendering and low memory usage across target devices.
- Implemented secure storage, background services, async programming, and Android lifecycle best practices for production-grade reliability.

PROJECTS

Tabyaan (تبيان) — All-in-One Islamic Companion App

2025 – Present

Technologies: Flutter, Dart, Firebase (Auth, Firestore, Storage), Hive, Bloc/Cubit, GetIt, GoRouter, Dartx, Dio, speech_to_text, video_player, LLMs model

- Solo-designed and built an all-in-one Islamic super-app end-to-end across 10+ feature modules — Quran reading, tafsir, hadith, fatwa Q&A, a live recitation coach, a zakat calculator, and a gamified prophetic-knowledge tree — using Clean Architecture (data/domain/presentation) throughout.
- Built a live Quran recitation checker using on-device speech recognition, matching spoken verses against Quran text word-by-word in real time with diacritic-normalized Arabic text comparison for accurate correct/incorrect feedback.

- Designed an offline-first reading experience with Hive-backed local persistence for Quran text, tafsir, hadith chapters, and reading history, keeping core features fully usable without a network connection.
- Built a gamified learning module (quizzes, leaderboard, rewards) to drive user engagement, plus an interactive Prophet lineage tree and battles timeline with custom-painted visualizations.
- Built a zakat calculator with a custom-painted scale visualization supporting multi-currency and gold/silver nisab calculations.
- Integrated Firebase Auth/Firestore/Storage for user accounts, fatwa submissions, and user-generated video content (create, save, and share short Islamic videos), plus full Arabic/English localization and light/dark theming.

Status: Preparing for Google Play launch. | GitHub: <https://github.com/MooHelmy/Tabyana>

Muadh (معاذ) — AI-Powered Content Protection App

Jan 2026 – May 2026

Technologies: Flutter, Dart, Kotlin, ONNX Runtime, Google ML Kit, Hive, Android VpnService, AccessibilityService, Device Admin API

- Architected and shipped an AI-powered Android content protection platform that detects and removes explicit media entirely on-device, with zero cloud dependency.
- Engineered a 3-model AI ensemble (NSFW classifier, face detector, skin analyzer) with confidence-based scoring and early-exit optimization, reducing processing overhead by ~25% while maintaining detection accuracy.
- Integrated Android AccessibilityService for real-time on-screen content monitoring across applications, with configurable alerts and policy enforcement.
- Built a real-time media monitoring pipeline using Android FileObserver to analyze newly created images and videos instantly, enabling continuous protection without manual scans.
- Implemented Device Admin security controls with encrypted PIN-based authorization, preventing unauthorized app removal or protection bypass.
- Developed a custom DNS-level filtering engine using Android VpnService to block explicit content across all browsers and apps, requiring no user configuration.
- Delivered a modular Clean Architecture build with Riverpod state management, Hive persistence, automated service recovery on reboot, and a real-time monitoring dashboard with per-source deletion analytics.

APKPure: apkpure.com/p/com.helmy.muadh | GitHub: github.com/MooHelmy/releases

ScanCare — AI Medical Diagnosis App (Graduation Project)

Dec 2023 – Jun 2024

Technologies: Flutter, Python, CNN, Firebase, Bloc, Google Maps API, Stripe

- Developed an AI-powered medical app that analyzes medical images using CNNs to detect health conditions with high diagnostic accuracy.
- Integrated 5 AI models, automating 70% of diagnostic tasks and significantly reducing average assessment time.
- Implemented Google Maps API for nearby clinic discovery, Stripe for secure payments, and encrypted patient-provider messaging.
- Built on Clean Architecture with Bloc state management, achieving a maintainable, fully testable codebase.

Demo: youtube.com/watch?v=ToaSB4hXt64

E-Commerce Backend API

Feb 2026

Technologies: ASP.NET Core Web API, Entity Framework Core, SQL Server, JWT Authentication

- Designed and built a RESTful backend API for an e-commerce platform using ASP.NET Core Web API and Entity Framework Core, with SQL Server as the data layer.
- Implemented JWT-based authentication and authorization to secure API endpoints.
- Structured the project using a clean, layered architecture for maintainability and testability.

GitHub: github.com/MooHelmy/.Net_Ecommerce_app

Flight Booking System — Airline Reservation REST API

Aug 2026 – Present

Technologies: ASP.NET Core (.NET 10), Entity Framework Core, SQL Server, ASP.NET Identity, JWT Authentication, Repository Pattern, OpenAPI/Scalar

- Architected a layered ASP.NET Core Web API (.NET 10) for airline flight booking, separating controllers, services, repositories, DTOs, and entities via dependency injection for a maintainable, testable codebase.
- Implemented JWT-based authentication and role-based authorization on ASP.NET Core Identity, with automated role and admin-account seeding at startup to secure all flight and booking endpoints.
- Built end-to-end booking workflows covering flight search, multi-tier fare classes (Economy, Business, First), and a seat-hold system that temporarily locks seats to prevent double-booking during checkout.
- Added a dedicated queue-management module for high-demand booking scenarios and a centralized global exception handler for consistent, standardized API error responses.
- Modeled the data layer with Entity Framework Core and SQL Server using code-first migrations, and documented all endpoints via OpenAPI/Scalar for interactive API exploration and testing.

GitHub: github.com/MooHelmy/Flight-booking-system

Hotel Management System — Hotel Operations REST API

Aug 2026 – Present

Technologies: ASP.NET Core (.NET 10), Entity Framework Core, SQL Server, ASP.NET Identity, JWT Authentication, REST API, Scalar

- Designed and built a full-featured ASP.NET Core Web API (.NET 10) for hotel operations, covering hotels, rooms, guests, employees, reservations, and payments across 7 dedicated controllers.
- Secured the API with JWT authentication and ASP.NET Identity, implementing role-based authorization and secure password management to protect all administrative and booking endpoints.
- Modeled a relational schema in SQL Server via Entity Framework Core linking hotels, rooms, amenities, guests, employees, reservations, and payments, with EF Core migrations managing schema evolution.
- Implemented reservation and check-in/check-out workflows that associate guests with rooms and track reservation and room-availability status in real time.
- Structured the codebase with a layered architecture (controllers, services, interfaces, DTOs, dependency injection) and documented the API with OpenAPI/Scalar for endpoint testing and exploration.

GitHub: github.com/MooHelmy/Hotel-Management-.net

EDUCATION

Bachelor of Computer Science | Benha University — Faculty of Computers and Artificial Intelligence, Computer Science Major
2020 – 2024

GPA: 3.3 | Graduation Project: Excellent

CERTIFICATIONS & TRAINING

- Flutter Developer — Information Technology Institute (ITI), Menoufia, Egypt (Jan – Apr 2023)
- Mobile Development using Flutter — ITCRC, Benha, Egypt (Jul – Oct 2023)
- Android Developer — ITCRC, Benha, Egypt (Jul – Oct 2022)

ADDITIONAL ACTIVITIES

Technical Content Creator | YouTube @eng_mo723

- Created 100+ educational videos covering Flutter, software engineering, embedded systems, Assembly programming, and Google Cloud.
- Built an engaged technical audience with 50,000+ total views across curated playlists.

SOFT SKILLS

Communication • Teamwork • Adaptability • Problem Solving • Time Management • Leadership